

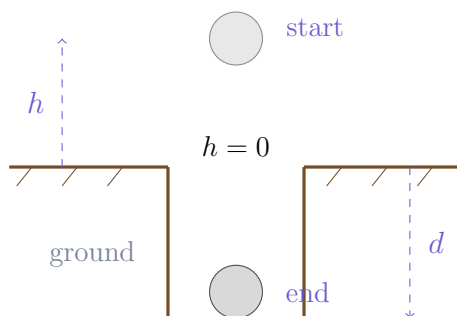
Electric Potential Energy (completed)

Goal.

- Build the idea of potential energy with gravity first: height above the ground, and what happens if you keep going *below* it.
- See the same idea for a charge pushed against a uniform field — for both a $+$ and a $-$ charge.
- Write the *full* (non-uniform) form of gravitational PE, and see that “higher” always means “more PE,” even though the value is negative.
- Use that same point-source structure to build the electric PE of two point charges, then a whole system of them.
- Combine two point charges’ PE to get the potential energy of a dipole in a field.

1. Gravitational PE: height, then a hole

A ball sits a height h above the ground. Ground level is our reference: $U = 0$ there.



$$U(h) = mgh$$

Now let the ball fall past ground level, into a hole of depth d .

$$U(\text{bottom}) = -mgd$$

Ground is still where $h = 0$. The bottom of the hole is *below* that reference, at height $-d$, so $U = mg(-d) = -mgd$ — our first negative PE.

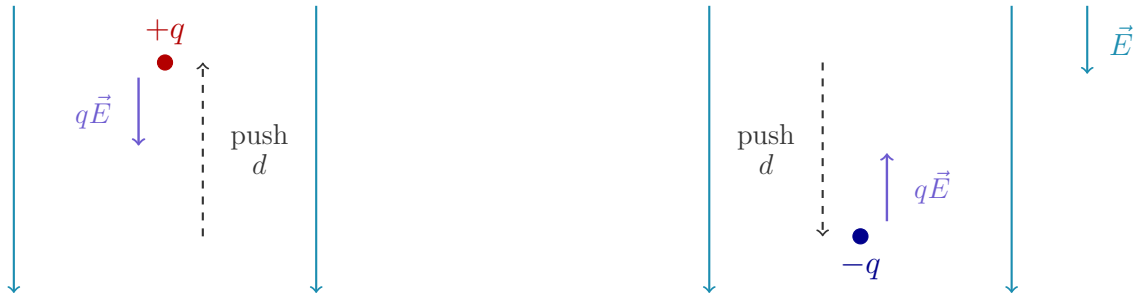
$$\Delta U = U(\text{bottom}) - U(h) = -mgd - mgh = -mg(h + d)$$

Check. The ball’s PE at the bottom of the hole came out negative. Is that a problem?

No — “zero” was just our choice of ground level. PE can absolutely be negative below wherever you chose to measure from. Keep this in mind: we’ll see the exact same thing with opposite charges.

2. The same idea, for a charge in a uniform field

Replace gravity with a uniform electric field $\vec{E}$ pointing *down* (same visual sense as gravity), and try it with both signs of charge.



- Force on the $+q$: $q\vec{E}$ points **downward**, the same direction as $\vec{E}$ — that's its *natural* (free) direction of motion.
- Force on the $-q$: $q\vec{E}$ points **upward**, opposite $\vec{E}$ — the sign flip turns the natural direction around.

Push each charge a distance d *against* its own natural direction — the same idea as raising the ball.

$$\Delta U_+ = qEd \qquad \Delta U_- = |q|Ed$$

- Both are **positive** — pushing *against* a charge's natural direction always stores positive PE, regardless of the charge's sign.
- Let either charge go instead, and it moves along its *natural* direction: PE **decreases** either way — the field does the work.

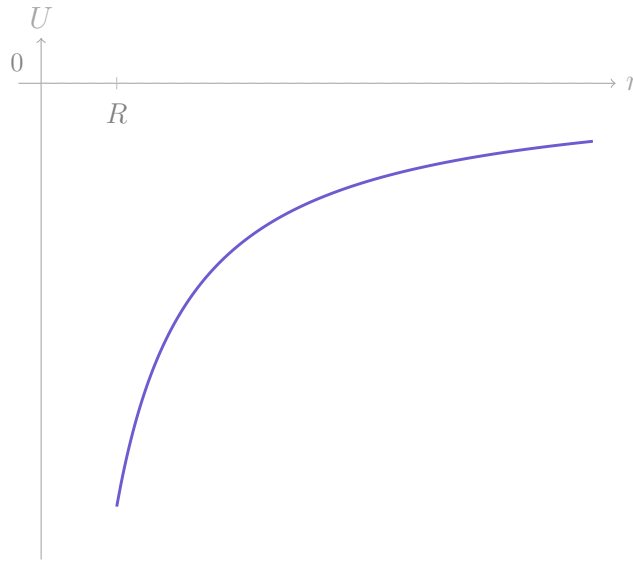
Check. A negative charge is released in this downward field. Which way does it move, and what happens to its PE?

It moves *upward* — opposite $\vec{E}$, since the force on a negative charge is $q\vec{E}$ with $q < 0$. Its PE decreases, same rule as always: released charges move toward lower PE.

3. Gravitational PE, the full (non-uniform) form

$U = mgh$ only works because g barely changes over any height you can reach near Earth's surface. Zoom out far enough to leave that “uniform field” approximation, and the real, general form (valid at any separation r between two masses) is:

$$U(r) = -\frac{GMm}{r}$$



- As $r \rightarrow \infty$, $U \rightarrow 0$ — same rule as always: zero PE at infinite separation.
- Every value of $U(r)$ here is **negative**. But moving to a *larger* r (further from Earth, “higher”) always makes U **less negative** — i.e. **larger, higher on the graph**. Don’t let the minus sign fool you: higher is *always* more PE, on this curve or any other.

Check. True or false: since $U(r)$ is always negative here, moving *further* from Earth makes U smaller.

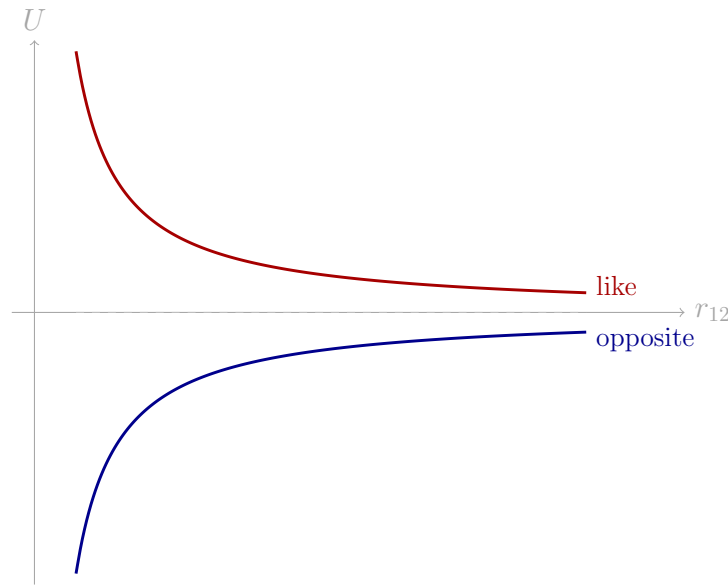
False. Further away makes U *larger* (less negative, closer to zero) — the graph is rising the whole way, it just never crosses above 0. “Negative” and “smaller” are not the same thing.

4. Electric PE of two point charges (from the class prep)

Pulling a pair of point charges apart, or pushing them together, takes work — work that gets stored as potential energy, exactly like the point-mass form above.

$$U(q_1, q_2) = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r_{12}}$$

Sketch U vs. r_{12} below, for an opposite-sign pair and a like-sign pair, on the same axes. (Same shape you just drew for gravity — plus its mirror image.)



- As $r_{12} \rightarrow \infty$, $U \rightarrow 0$.
- For three or more charges, sum over every **pair** (not every charge).
- Opposite charges (+ and -): U is **negative** at any finite separation, and $U \rightarrow 0$ only as $r \rightarrow \infty$. So $U = 0$ is actually the **largest** value here — pulling the charges closer always finds a *lower* (more negative) PE.
- Like charges (same sign): U is **positive** at any finite separation, and again $U \rightarrow 0$ as $r \rightarrow \infty$. Here $U = 0$ is the **smallest** value — you can't push like charges to a lower PE than infinitely far apart.

$$U_{\text{total}} = \frac{1}{4\pi\epsilon_0} \sum_{\text{pairs } (i,j)} \frac{q_i q_j}{r_{ij}}$$

Check. Two $+q$ charges are brought from far apart to a separation r . Is U positive or negative? What about a $+q$ and a $-q$?

$+q$ and $+q$: positive (repulsive, energy stored pushing them together). $+q$ and $-q$: negative (attractive) — just like the ball in the hole.

5. The dipole in a field

A dipole is a $+q$ and a $-q$ separated by $\vec{d}$ (dipole moment $\vec{p} = q\vec{d}$), sitting in a uniform field $\vec{E}$, tilted at angle θ between $\vec{p}$ and $\vec{E}$. Each charge sits at a different place along $\vec{E}$ — use §2's pushing idea on each charge, then add.

Split $\vec{d}$ into a component along $\vec{E}$ ($d \cos \theta$) and a component across it. Only the along- $\vec{E}$ part matters (§2): the $+q$ sits $\frac{1}{2}d \cos \theta$ further along $\vec{E}$ than center, the $-q$ sits $\frac{1}{2}d \cos \theta$ behind it. Adding the two charges' PE (each $\propto -qEd$, from §2, since moving *with* $\vec{E}$ lowers PE) gives $U = -qEd \cos \theta = -\vec{p} \cdot \vec{E}$.

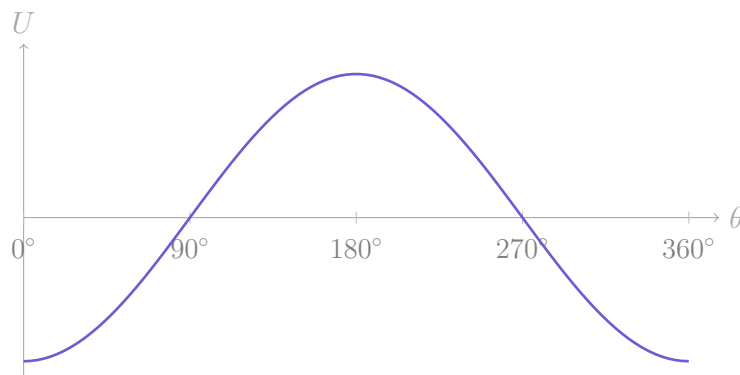
$$U(\theta) = -pE \cos \theta$$

- At $\theta = 90^\circ$ (perpendicular to the field): $U = 0$.
- At $\theta = 0^\circ$ (aligned with the field): $U = -pE$ — which is **lower** than the $\theta = 90^\circ$ value!
- At $\theta = 180^\circ$ (anti-aligned): $U = +pE$.

Check. Which orientation is the most stable (lowest PE)? Is that the orientation where $U = 0$? What does this have in common with the ball in the hole?

$\theta = 0^\circ$ (aligned) is most stable, at $U = -pE$ — *not* the $U = 0$ orientation. Just like the ball in the hole: $U = 0$ is just wherever we started measuring from, not necessarily the lowest energy the system can reach.

6. Sketch it: U vs. θ



Check your curves, and the point-charge and uniform-field results above, against the *Assembling Charges*, *PE vs. Separation*, and *Electric Dipole in a Field* sims — open their **Info** panels.